

Configuraciones de equipos de interconexión

Guía para configurar desde cero los equipos OMI de la topología de Cisco Packet Tracer. Cada bloque IOS incluye hostname, interfaces, VLAN, enrutamiento o seguridad según corresponda, contraseñas y guardado.

Credenciales

Uso	Contraseña
Acceso por consola	ConsolaOMI
Modo privilegiado enable	OMI2026
Acceso VTY	VTYOMI

Router Alumnos

Conexiones:

- Gi0/0 a DIST-ALUMNOS Gi0/1
- Gi0/1 a Frontera Gi0/0

```
None
enable
configure terminal
hostname Alumnos
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #

line console 0
password ConsolaOMI
login
logging synchronous
exit

line vty 0 4
password VTYOMI
login
exit

interface GigabitEthernet0/0
description LAN CONCURSANTES HACIA DIST-ALUMNOS
ip address 172.23.24.1 255.255.254.0
ip helper-address 172.23.28.226
```

```
no shutdown
exit

interface GigabitEthernet0/1
description TRANSITO HACIA FRONTERA
ip address 172.23.28.233 255.255.255.252
no shutdown
exit

router eigrp 777
network 172.23.24.0 0.0.1.255
network 172.23.28.232 0.0.0.3
no auto-summary
exit

ip route 0.0.0.0 0.0.0.0 172.23.28.234
end
write memory
```

Router Frontera

Conexiones:

- Gi0/0 a Alumnos Gi0/1
- Gi0/1 a SWDistribucionExternos Gi0/1
- Gi0/2 a FronteraCampus Gi0/0

```
None

enable
configure terminal
hostname Frontera
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #

line console 0
password ConsolaOMI
login
logging synchronous
exit

line vty 0 4
password VTYOMI
login
exit

interface GigabitEthernet0/0
description TRANSITO HACIA ALUMNOS
ip address 172.23.28.234 255.255.255.252
```

```
ip nat inside
no shutdown
exit

interface GigabitEthernet0/1
description TRONCAL HACIA SWDistribucionExternos
no ip address
no shutdown
exit

interface GigabitEthernet0/1.10
description JUECES
encapsulation dot1Q 10
ip address 172.23.28.1 255.255.255.224
ip helper-address 172.23.28.226
ip nat inside
exit

interface GigabitEthernet0/1.20
description ENTRENADORES
encapsulation dot1Q 20
ip address 172.23.28.65 255.255.255.192
ip helper-address 172.23.28.226
ip nat inside
exit

interface GigabitEthernet0/1.30
description PRENSA
encapsulation dot1Q 30
ip address 172.23.28.129 255.255.255.192
ip helper-address 172.23.28.226
ip nat inside
exit

interface GigabitEthernet0/1.40
description INVITADOS
encapsulation dot1Q 40
ip address 172.23.26.1 255.255.254.0
ip helper-address 172.23.28.226
ip nat inside
exit

interface GigabitEthernet0/1.60
description INFRAESTRUCTURA
encapsulation dot1Q 60
ip address 172.23.28.193 255.255.255.224
ip helper-address 172.23.28.226
ip nat inside
exit

interface GigabitEthernet0/1.70
```

```

description SERVIDORES
encapsulation dot1Q 70
ip address 172.23.28.225 255.255.255.248
ip nat inside
exit

interface GigabitEthernet0/2
description SALIDA HACIA CAMPUS
ip address 10.32.100.100 255.255.255.0
ip nat outside
no shutdown
exit

router eigrp 777
network 172.23.24.0 0.0.7.255
no auto-summary
exit

access-list 7 permit 172.23.24.0 0.0.7.255
ip nat inside source list 7 interface GigabitEthernet0/2 overload
ip route 0.0.0.0 0.0.0.0 10.32.100.1

ip access-list extended BLOQUEAR-CONCURSANTES
deny ip 172.23.26.0 0.0.1.255 172.23.24.0 0.0.1.255
deny ip 172.23.28.64 0.0.0.63 172.23.24.0 0.0.1.255
deny ip 172.23.28.128 0.0.0.63 172.23.24.0 0.0.1.255
permit ip any any
exit

interface GigabitEthernet0/0
ip access-group BLOQUEAR-CONCURSANTES out
exit

end
write memory

```

SWDistribucionExternos

Mapa de puertos:

Puerto	Destino	VLAN
Fa0/1	SW-ENH-P1-PRENSA Fa0/24	30
Fa0/2	AP-ENH-P1-PRENSA Port 0	30
Fa0/10	SW-CIT-P1-ENTRENADORES Fa0/24	20
Fa0/11	AP-CIT-P1-ENTRENADORES Port 0	20

Fa0/15	SW-EIC-P1-JUECES Fa0/24	10
Fa0/20	AP-ENH-P1-INVITADOS Port 0	40
Fa0/21-22	AP adicionales de invitados	40
Fa0/24	DHCP-DNS-OMI Fa0	70
Gi0/1	Frontera Gi0/1	Troncal

```

None
enable
configure terminal
hostname SWDistribucionExternos
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #

line console 0
password ConsolaOMI
login
logging synchronous
exit

line vty 0 4
password VTYOMI
login
exit

vlan 10
name JUECES
vlan 20
name ENTRENADORES
vlan 30
name PRENSA
vlan 40
name INVITADOS
vlan 60
name INFRAESTRUCTURA
vlan 70
name SERVIDORES

interface range FastEthernet0/1 - 2
switchport mode access
switchport access vlan 30
spanning-tree portfast
exit

interface range FastEthernet0/10 - 11

```

```

switchport mode access
switchport access vlan 20
spanning-tree portfast
exit

interface FastEthernet0/15
switchport mode access
switchport access vlan 10
spanning-tree portfast
exit

interface range FastEthernet0/20 - 22
switchport mode access
switchport access vlan 40
spanning-tree portfast
exit

interface FastEthernet0/24
switchport mode access
switchport access vlan 70
spanning-tree portfast
exit

interface GigabitEthernet0/1
description TRONCAL HACIA FRONTERA
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,60,70
exit

end
write memory

```

DIST-ALUMNOS

Mapa de puertos:

Puerto	Destino
Fa0/1	SW-CIT-P3-CONG-04 Fa0/24
Fa0/2	SW-CIT-P3-CONG-05 Fa0/24
Fa0/3	SW-ALUMNOS-RESERVA-01 Fa0/24
Fa0/4	SW-CIT-P3-CONG-03 Fa0/24
Fa0/5	SW-CIT-P3-CONG-01 Fa0/24

Fa0/6	SW-CIT-P3-CONG-02 Fa0/24
Fa0/7	SW-ALUMNOS-RESERVA-02 Fa0/24
Fa0/8	SW-ENH-P2-1223 Fa0/24
Fa0/9	SW-ENH-P2-1224 Fa0/24
Fa0/10	SW-EIC-P1-12102 Fa0/24
Fa0/11	SW-ENH-P2-FIN Fa0/24
Gi0/1	Alumnos Gi0/0

```
None
enable
configure terminal
hostname DIST-ALUMNOS
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #

line console 0
password ConsolaOMI
login
logging synchronous
exit

line vty 0 4
password VTYOMI
login
exit

interface range FastEthernet0/1 - 11
switchport mode access
switchport access vlan 1
exit

interface GigabitEthernet0/1
description HACIA ROUTER ALUMNOS
switchport mode access
switchport access vlan 1
exit

end
write memory
```

Switches de acceso para concursantes

En cada switch, Fa0/24 es el uplink hacia DIST-ALUMNOS. Los puertos Fa0/1-23 son puertos de terminales concursantes en VLAN 1.

SW-CIT-P3-CONG-01

```
None
enable
configure terminal
hostname SW-CIT-P3-CONG-01
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
  logging synchronous
  exit
line vty 0 4
  password VTYOMI
  login
  exit
interface range FastEthernet0/1 - 23
  switchport mode access
  switchport access vlan 1
  spanning-tree portfast
  exit
interface FastEthernet0/24
  description UPLINK DIST-ALUMNOS FA0/5
  switchport mode access
  switchport access vlan 1
  exit
end
write memory
```

SW-CIT-P3-CONG-02

```
None
enable
configure terminal
hostname SW-CIT-P3-CONG-02
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
```



```
logging synchronous
exit
line vty 0 4
password VTYOMI
login
exit
interface range FastEthernet0/1 - 23
switchport mode access
switchport access vlan 1
spanning-tree portfast
exit
interface FastEthernet0/24
description UPLINK DIST-ALUMNOS FA0/6
switchport mode access
switchport access vlan 1
exit
end
write memory
```

SW-CIT-P3-CONG-03

```
None
enable
configure terminal
hostname SW-CIT-P3-CONG-03
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
password ConsolaOMI
login
logging synchronous
exit
line vty 0 4
password VTYOMI
login
exit
interface range FastEthernet0/1 - 23
switchport mode access
switchport access vlan 1
spanning-tree portfast
exit
interface FastEthernet0/24
description UPLINK DIST-ALUMNOS FA0/4
switchport mode access
switchport access vlan 1
exit
end
write memory
```

SW-CIT-P3-CONG-04

```
None
enable
configure terminal
hostname SW-CIT-P3-CONG-04
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
  logging synchronous
  exit
line vty 0 4
  password VTYOMI
  login
  exit
interface range FastEthernet0/1 - 23
  switchport mode access
  switchport access vlan 1
  spanning-tree portfast
  exit
interface FastEthernet0/24
  description UPLINK DIST-ALUMNOS FA0/1
  switchport mode access
  switchport access vlan 1
  exit
end
write memory
```

SW-CIT-P3-CONG-05

```
None
enable
configure terminal
hostname SW-CIT-P3-CONG-05
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
  logging synchronous
  exit
line vty 0 4
  password VTYOMI
  login
  exit
interface range FastEthernet0/1 - 23
```

```
switchport mode access
switchport access vlan 1
spanning-tree portfast
exit
interface FastEthernet0/24
description UPLINK DIST-ALUMNOS FA0/2
switchport mode access
switchport access vlan 1
exit
end
write memory
```

SW-ALUMNOS-RESERVA-01

```
None
enable
configure terminal
hostname SW-ALUMNOS-RESERVA-01
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
password ConsolaOMI
login
logging synchronous
exit
line vty 0 4
password VTYOMI
login
exit
interface range FastEthernet0/1 - 23
switchport mode access
switchport access vlan 1
spanning-tree portfast
exit
interface FastEthernet0/24
description UPLINK DIST-ALUMNOS FA0/3
switchport mode access
switchport access vlan 1
exit
end
write memory
```

SW-ALUMNOS-RESERVA-02

```
None
enable
configure terminal
hostname SW-ALUMNOS-RESERVA-02
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
  logging synchronous
  exit
line vty 0 4
  password VTYOMI
  login
  exit
interface range FastEthernet0/1 - 23
  switchport mode access
  switchport access vlan 1
  spanning-tree portfast
  exit
interface FastEthernet0/24
  description UPLINK DIST-ALUMNOS FA0/7
  switchport mode access
  switchport access vlan 1
  exit
end
write memory
```

SW-ENH-P2-1223

```
None
enable
configure terminal
hostname SW-ENH-P2-1223
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
  logging synchronous
  exit
line vty 0 4
  password VTYOMI
  login
  exit
interface range FastEthernet0/1 - 23
```

```
switchport mode access
switchport access vlan 1
spanning-tree portfast
exit
interface FastEthernet0/24
description UPLINK DIST-ALUMNOS FA0/8
switchport mode access
switchport access vlan 1
exit
end
write memory
```

SW-ENH-P2-1224

```
None
enable
configure terminal
hostname SW-ENH-P2-1224
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
password ConsolaOMI
login
logging synchronous
exit
line vty 0 4
password VTYOMI
login
exit
interface range FastEthernet0/1 - 23
switchport mode access
switchport access vlan 1
spanning-tree portfast
exit
interface FastEthernet0/24
description UPLINK DIST-ALUMNOS FA0/9
switchport mode access
switchport access vlan 1
exit
end
write memory
```

SW-ENH-P2-FIN

```
None
enable
configure terminal
hostname SW-ENH-P2-FIN
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
  logging synchronous
  exit
line vty 0 4
  password VTYOMI
  login
  exit
interface range FastEthernet0/1 - 23
  switchport mode access
  switchport access vlan 1
  spanning-tree portfast
  exit
interface FastEthernet0/24
  description UPLINK DIST-ALUMNOS FA0/11
  switchport mode access
  switchport access vlan 1
  exit
end
write memory
```

SW-EIC-P1-12102

```
None
enable
configure terminal
hostname SW-EIC-P1-12102
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
  logging synchronous
  exit
line vty 0 4
  password VTYOMI
  login
  exit
interface range FastEthernet0/1 - 23
```

```
switchport mode access
switchport access vlan 1
spanning-tree portfast
exit
interface FastEthernet0/24
description UPLINK DIST-ALUMNOS FA0/10
switchport mode access
switchport access vlan 1
exit
end
write memory
```

Switches de segmentos externos

SW-EIC-P1-JUECES

Fa0/24 conecta a SWDistribucionExternos Fa0/15. Los jueces se conectan por cable a Fa0/1-23

```
None
enable
configure terminal
hostname SW-EIC-P1-JUECES
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
password ConsolaOMI
login
logging synchronous
exit
line vty 0 4
password VTYOMI
login
exit
vlan 10
name JUECES
interface range FastEthernet0/1 - 23
switchport mode access
switchport access vlan 10
spanning-tree portfast
exit
interface FastEthernet0/24
description UPLINK SWDistribucionExternos FA0/15
switchport mode access
switchport access vlan 10
exit
end
```

```
write memory
```

SW-CIT-P1-ENTRENADORES

Fa0/24 conecta a SWDistribucionExternos Fa0/10

```
None
enable
configure terminal
hostname SW-CIT-P1-ENTRENADORES
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
line console 0
  password ConsolaOMI
  login
  logging synchronous
  exit
line vty 0 4
  password VTYOMI
  login
  exit
vlan 20
  name ENTRENADORES
interface range FastEthernet0/1 - 23
  switchport mode access
  switchport access vlan 20
  spanning-tree portfast
  exit
interface FastEthernet0/24
  description UPLINK SWDistribucionExternos FA0/10
  switchport mode access
  switchport access vlan 20
  exit
end
write memory
```

SW-ENH-P1-PRENSA

Fa0/24 conecta a SWDistribucionExternos Fa0/1

```
None
enable
configure terminal
hostname SW-ENH-P1-PRENSA
enable secret OMI2026
service password-encryption
banner motd # ACCESO EXCLUSIVO PERSONAL AUTORIZADO OMI #
```



```
line console 0
 password ConsolaOMI
 login
 logging synchronous
 exit
line vty 0 4
 password VTYOMI
 login
 exit
vlan 30
 name PRENSA
interface range FastEthernet0/1 - 23
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 exit
interface FastEthernet0/24
 description UPLINK SWDistribucionExternos FA0/1
 switchport mode access
 switchport access vlan 30
 exit
end
write memory
```

Access Points

AP-CIT-P1-ENTRENADORES

1. Conectar Port 0 a SWDistribucionExternos Fa0/11 con Copper Straight-Through.
2. Abrir el AP y entrar a Config > Wireless.
3. Configurar SSID: OMI-ENTRENADORES.
4. Seleccionar WPA2-PSK.
5. Configurar la clave EntrenaOMI2026.

AP-ENH-P1-PRENSA

1. Conectar Port 0 a SWDistribucionExternos Fa0/2 con Copper Straight-Through.
2. Abrir el AP y entrar a Config > Wireless.
3. Configurar SSID: OMI-PRENSA.
4. Seleccionar WPA2-PSK.
5. Configurar la clave PrensaOMI2026.

AP-ENH-P1-INVITADOS

1. Conectar Port 0 a SWDistribucionExternos Fa0/20 con Copper Straight-Through.

2. Abrir el AP y entrar a Config > Wireless.
3. Configurar SSID: OMI-INVITADOS.
4. Seleccionar WPA2-PSK.
5. Configurar la clave InvitadosOMI2026.
6. En cada laptop invitada, seleccionar OMI-INVITADOS, escribir la clave y elegir DHCP.

Servidor DHCP-DNS-OMI

Interfaz FastEthernet0

En Desktop > IP Configuration > Static:

Campo	Valor
IP Address	172.23.28.226
Subnet Mask	255.255.255.248
Default Gateway	172.23.28.225
DNS Server	172.23.28.226

Conectar FastEthernet0 a SWDistribucionExternos Fa0/24 con Copper Straight-Through.

DHCP

Entrar a Services > DHCP, seleccionar On y crear todos los pools:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User
JUECES	172.23.28.1	172.23.28.226	172.23.28.10	255.255.25...	20
INFRAESTRUCTURA	172.23.28.193	172.23.28.226	172.23.28.200	255.255.25...	20
PRENSA	172.23.28.129	172.23.28.226	172.23.28.135	255.255.25...	32
ENTRENADORES	172.23.28.65	172.23.28.226	172.23.28.70	255.255.25...	40
INVITADOS	172.23.26.1	172.23.28.226	172.23.26.10	255.255.254.0	300
CONCURSANTES	172.23.24.1	172.23.28.226	172.23.24.10	255.255.254.0	330

DNS

Entrar a Services > DNS, seleccionar On y agregar:

Nombre	Dirección
servicios.omi.local	172.23.28.226
concurso.omi.local	200.1.1.50

Página web interna

Entrar a Services > HTTP, seleccionar HTTP: On, editar index.html y guardar:

HTML

```
<html>
<head><title>Servicios OMI</title></head>
<body>
  <h1>Servicios internos de la XXV OMI</h1>
  <p>Servidor DHCP, DNS y servicios internos disponible.</p>
</body>
</html>
```

Servidor Concurso

1. Abrir Concurso y entrar a Desktop > IP Configuration > Static.
2. Configurar IP Address: 200.1.1.50.
3. Configurar Subnet Mask: 255.255.255.0.
4. Configurar Default Gateway: 200.1.1.1.
5. Entrar a Services > HTTP y seleccionar HTTP: On.
6. Editar index.html y guardar:

HTML

```
<html>
<head><title>XXV OMI</title></head>
<body>
  <h1>XXV Olimpiada Mexicana de Informática</h1>
  <p>Portal oficial del concurso.</p>
</body>
</html>
```

Terminales

En cada PC o laptop cableada:

1. Abrir Desktop > IP Configuration.
2. Seleccionar DHCP.
3. Confirmar dirección, máscara, gateway y DNS del segmento.

Terminales concursantes y switch:

Terminal	Switch	Puerto
PC-CIT-CONG-01	SW-CIT-P3-CONG-01	Fa0/1
PC-CIT-CONG-02	SW-CIT-P3-CONG-02	Fa0/1
PC-CIT-CONG-03	SW-CIT-P3-CONG-03	Fa0/1
PC-CIT-CONG-04	SW-CIT-P3-CONG-04	Fa0/1
PC-CIT-CONG-05	SW-CIT-P3-CONG-05	Fa0/1
PC-ALUMNOS-RESERVA-01	SW-ALUMNOS-RESERVA-01	Primer puerto libre
PC-ALUMNOS-RESERVA-02	SW-ALUMNOS-RESERVA-02	Primer puerto libre
PC-ENH-1223	SW-ENH-P2-1223	Fa0/1
PC-ENH-1224	SW-ENH-P2-1224	Fa0/1
PC-ENH-FIN	SW-ENH-P2-FIN	Fa0/1
PC-EIC-12102	SW-EIC-P1-12102	Fa0/1

Terminales externos:

Terminal	Equipo de acceso	Puerto / medio
PC-EIC-JUEZ	SW-EIC-P1-JUECES	Fa0/1
PC-CIT-ENTRENADOR	SW-CIT-P1-ENTRENADOR ES	Fa0/1
PC-ENH-PRENSA	SW-ENH-P1-PRENSA	Fa0/1
Laptop-ENH-INVITADO	AP-ENH-P1-INVITADOS	Inalámbrico